

UBCx Phot1x — Silicon Photonics Design, Fabrication and Data Analysis

Mach-Zehnder Interferometer Final Design Report

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Device: MZI1 — TE Polarisation @ 1550 nm

Fabrication Run: openEBL-2026-05 | ANT EBeam Process

GitHub: github.com/Mishuk1995/openEBL-2026-05

Date: 6th July 2026

Abstract—This report presents the design, simulation, layout, fabrication and measurement analysis of a TE-polarised Mach-Zehnder Interferometer (MZI) on a silicon-on-insulator (SOI) platform. The device was designed using KLayout with the SiEPIC-EBeam PDK and fabricated via electron-beam lithography through the openEBL-2026-05 community run using the Applied Nanotools (ANT) EBeam process. The MZI employs a 500×220 nm strip waveguide with a path length difference of $\Delta L = 56.397$ μm . Lumerical MODE gives a TE group index of $n_g = 4.195887$ at 1550 nm, predicting an FSR of 10.15 nm. The fabricated chip returns an average FSR of 10.062 nm and $n_g = 4.234$ within 1% of simulation, though the lowest-CV single reading and the average both fall slightly (<1%) outside the nominal corner analysis bounds. All measurement details and the resulting bound discussion are given in Section VII.

Keywords—silicon photonics, Mach-Zehnder interferometer, SOI, TE polarisation, free spectral range, electron-beam lithography, Lumerical MODE

I. INTRODUCTION

Silicon-on-insulator (SOI) is one of the most widely used platforms for integrated photonics. The large refractive index contrast between the silicon core ($n \approx 3.48$) and the surrounding silicon dioxide cladding ($n \approx 1.44$) gives tight optical confinement, which is why devices can be made very compact [1], [3].

The Mach-Zehnder Interferometer (MZI) is a straightforward device to analyse. Light is split into two waveguide arms of different length, picks up a wavelength-dependent phase difference, and is then recombined. The interference at the output gives a sinusoidal transmission spectrum whose fringe spacing (the free spectral range (FSR)) depends on the arm length difference and the waveguide group index [2].

This report focuses on the TE polarisation mode at a design wavelength of 1550 nm. The device was submitted to the openEBL-2026-05 community fabrication run and characterised using automated optical probing. Two questions are addressed: how well does Lumerical MODE predict the measured FSR and group index, and does the fabricated device fall within the performance range expected from ANT EBeam process tolerances?

II. THEORY

A. MZI Transfer Function

An imbalanced MZI with a 50/50 splitter and arm length difference ΔL produces the following transmission at output port 1 [1]:

$$T(\lambda) = 1/2 [1 + \cos(\Delta\phi(\lambda))] \quad (1)$$

where the optical phase difference between the two arms is:

$$\Delta\phi(\lambda) = (2\pi/\lambda) n_{\text{eff}}(\lambda) \Delta L \quad (2)$$

and $n_{\text{eff}}(\lambda)$ is the wavelength-dependent effective index of the guided mode. Transmission at output port 2 is the complement: $T_2(\lambda) = 1 - T_1(\lambda)$.

B. Free Spectral Range

The spacing between consecutive transmission maxima, the free spectral range (FSR), is:

$$\text{FSR} = \lambda^2/(n_g \Delta L) \quad (3)$$

where λ is the operating wavelength and n_g is the group index, defined as:

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda \cdot d n_{\text{eff}}/d\lambda \quad (4)$$

The group index captures waveguide dispersion and directly governs the FSR. For TE-polarised light in a 500×220 nm SOI strip waveguide, the TE mode has stronger optical confinement than the TM mode, resulting in a higher effective index and a correspondingly higher group index.

III. DEVICE DESIGN & LAYOUT

The MZI was designed using KLayout (version 0.28+) with the SiEPIC-EBeam PDK. All components were selected from verified PDK cells to ensure compliance with ANT EBeam design rules. The arm lengths were read directly from the path-length values encoded in the SiEPIC PDK waveguide cell names within the GDS file EBeam_ndcmishuk_TE.gds.

The circuit uses a fibre grating coupler for optical input, an ebeam_y_1550 Y-branch as the 50/50 splitter, two waveguide arms of different length, an ebeam_bdc_te1550 broadband directional coupler as the combiner, and output grating couplers routed to Detectors 1–4 in the automated test (Detectors 1 and 4 recorded no signal above the noise floor in either measurement session; see Section VII).

TABLE I. MZII DEVICE DESIGN PARAMETERS. GREEN CELLS ARE VALUES CONFIRMED FROM THE GDS FILE.

Parameter	Value	Unit	Source
Path length difference (ΔL)	56.397	μm	KLayout GDS: EBeam_ndcmishuk_TE.gds
Arm 1 length (L_1)	53.248	μm	KLayout GDS: EBeam_ndcmishuk_TE.gds
Arm 2 length (L_2)	109.645	μm	KLayout GDS: EBeam_ndcmishuk_TE.gds
Waveguide width (nominal)	500	nm	SiEPIC-EBeam PDK specification
Waveguide height (nominal)	220	nm	ANT EBeam process specification
Polarisation	TE	—	Design intent; confirmed by measurement (Section VII)
Design wavelength	1550	nm	Design intent
Splitter	ebeam_y_1550	—	SiEPIC-EBeam PDK
Combiner	ebeam_bdc_te1550	—	SiEPIC-EBeam PDK
Min. feature size	60	nm	ANT EBeam process constraint
Fabrication run	openEBL-2026-05	—	Community fabrication submission

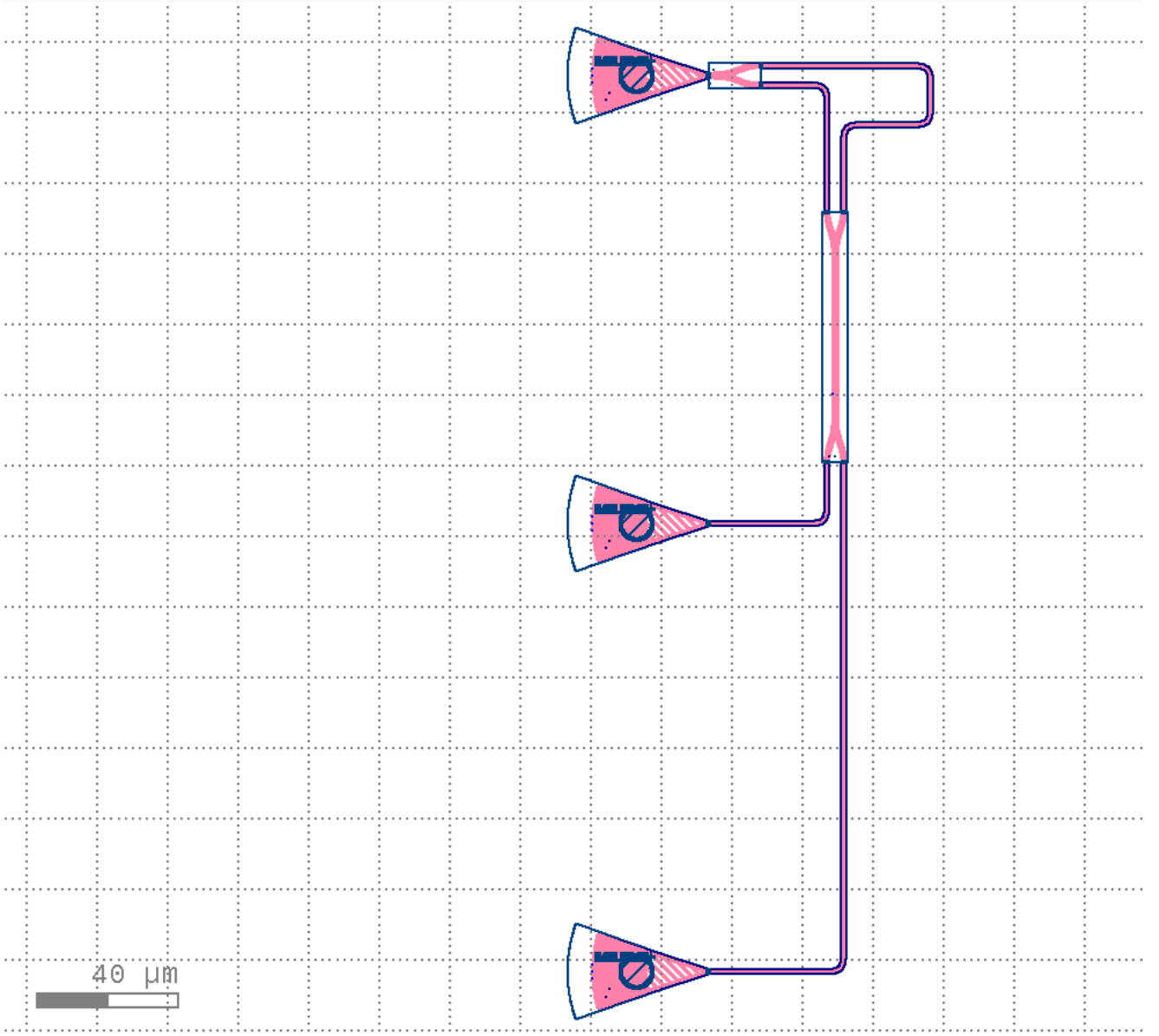


Fig. 1. KLayout screenshot of the TE MZII layout. Input grating coupler — confirmed as the input via its GDS pin label (*opt1in*) and via SiEPIC-Tools' INTERCONNECT auto-routing, which wires this port to the laser source — Y-branch splitter, two waveguide arms of unequal length ($\Delta L = 56.397 \mu\text{m}$), broadband directional coupler combiner, and output grating couplers.

IV. WAVEGUIDE SIMULATION

The fundamental TE mode (Mode 1) of the $500 \times 220 \text{ nm}$ SOI strip waveguide was simulated using the Lumerical MODE finite-difference eigenmode (FDE) solver at $\lambda = 1550 \text{ nm}$. The TE mode was identified as the mode with the

dominant E_x field component and the highest TE polarisation fraction (98%) among the guided modes.

The effective index $n_{\text{eff}} = 2.444191$ and group index $n_g = 4.195887$ were extracted at the nominal waveguide geometry. The group index was computed using the built-in dispersion analysis (Eq. 4). These two values are the inputs to every FSR prediction in this report.

TABLE II. LUMERICAL MODE SIMULATION RESULTS FOR TE MODE AT 1550 NM. GREEN CELLS ARE SIMULATION OUTPUTS USED THROUGHOUT THIS REPORT.

Parameter	Value	Unit	Source
Effective index n_{eff}	2.444191	—	Lumerical MODE, TE Mode 1, $\lambda=1550$ nm
Group index n_g	4.195887	—	Lumerical MODE, TE Mode 1, $\lambda=1550$ nm
Mode	TE (Mode 1)	—	Highest TE fraction (98%)
Waveguide width	500	nm	Nominal design (PDK)
Waveguide height	220	nm	ANT process nominal
Cladding material	SiO ₂	—	ANT EBeam process (buried oxide + top oxide)
Simulation region	2.5×2.0	μm	Course-recommended FDE settings

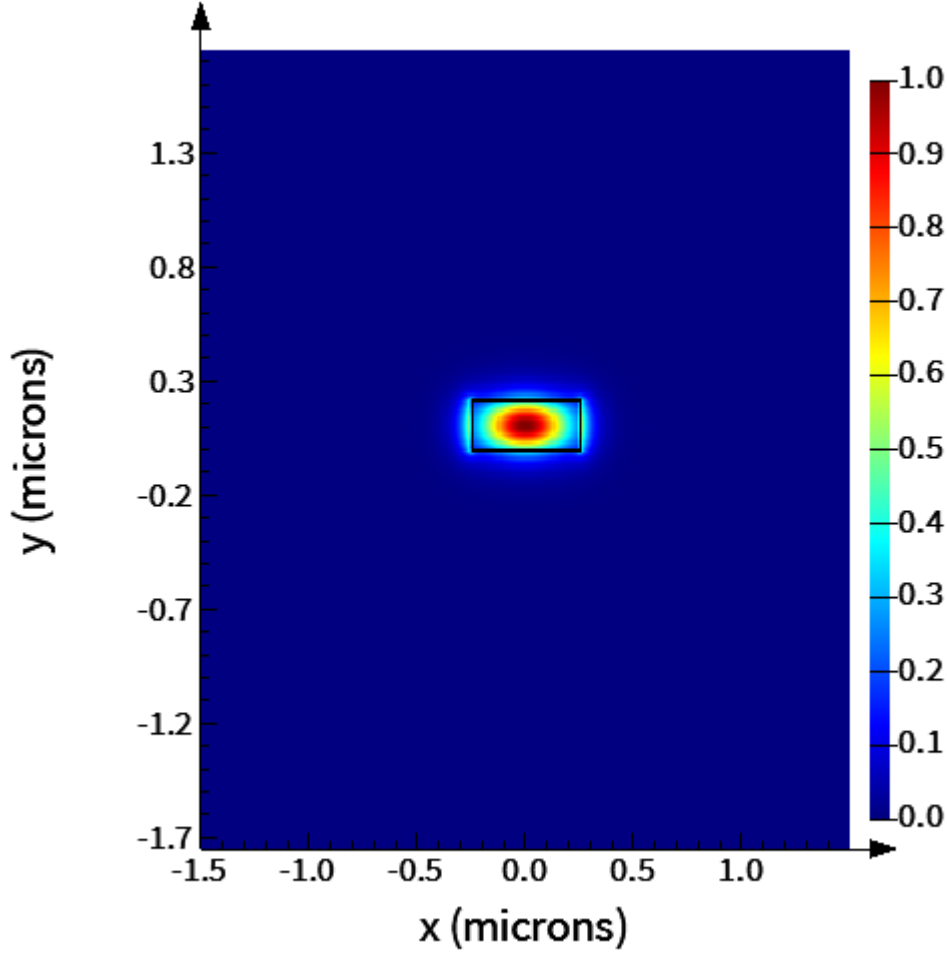


Fig. 2. TE mode field profile (E_x dominant component) in the 500×220 nm SOI strip waveguide at 1550 nm, from Lumerical MODE. The mode is identified as Mode 1 with TE polarisation fraction 98%.

V. MZI TRANSMISSION SIMULATION

The MZI transmission spectrum was calculated using the analytical transfer function (Eqs. 1–3) with parameters from Lumerical MODE. Using $\Delta L = 56.397 \mu\text{m}$ and $n_{\text{eff}}(\lambda)$ built from $n_1 = 2.444191$, $n_2 = (n_1 - n_g)/\lambda_0 = -1.130127 \mu\text{m}^{-1}$:

$$\text{FSR} = (1550 \text{ nm})^2 / (4.195887 \times 56397 \text{ nm}) = 10.153 \text{ nm} \approx 10.15 \text{ nm} \quad (5)$$

The calculation runs from 1500 to 1600 nm, matching the ANT automated test sweep range.

Note: The transmission spectrum above uses the analytical transfer function (Eq. 1) with a first-order Taylor expansion of $n_{\text{eff}}(\lambda)$ about $\lambda_0 = 1550$ nm. A Lumerical INTERCONNECT circuit-level simulation using the SiEPIC-EBeam compact model library was also performed as an independent cross-check; see the INTERCONNECT validation figure later in this section.

TABLE III. MZI SIMULATION PARAMETERS AND RESULTS. GREEN CELLS ARE THE KEY OUTPUTS USED IN THE MEASUREMENT COMPARISON (SECTION VII).

Parameter	Value	Unit	Source
Path length difference ΔL	56.397	μm	KLayout GDS: EBeam ndcmishuk TE.gds
Group index n_g	4.195887	—	Lumerical MODE, TE, $\lambda=1550$ nm
Effective index n_{eff}	2.444191	—	Lumerical MODE, TE, $\lambda=1550$ nm
Design wavelength λ	1550	nm	Design intent
Predicted FSR	10.15	nm	Calculated: $\lambda^2 / (n_g \times \Delta L)$, Eq. 3
Simulation method	Analytical (Eq. 1)	—	Python; INTERCONNECT confirms
Wavelength range	1500–1600	nm	ANT automated test sweep range

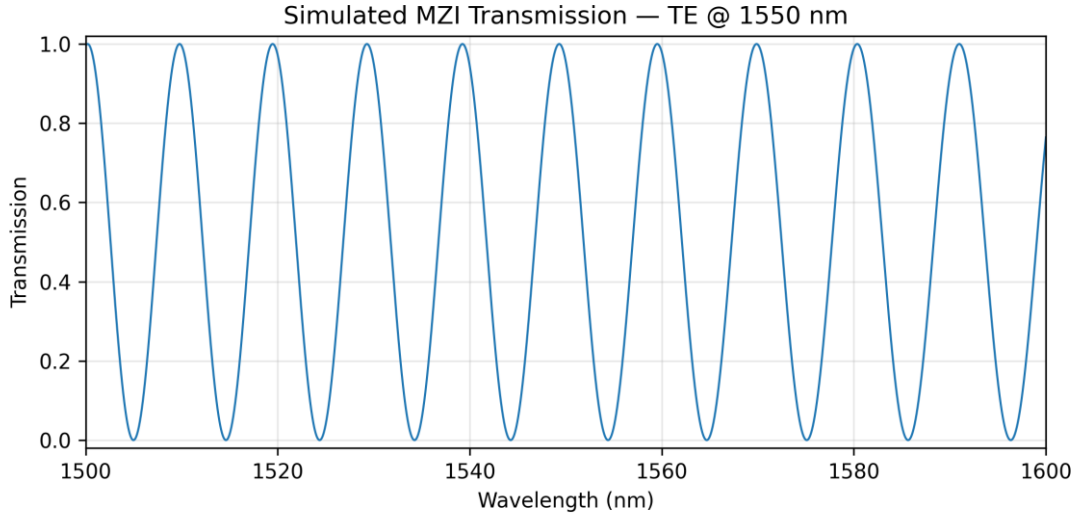


Fig. 3. Simulated MZI transmission spectrum (Eq. 1), $\lambda = 1500\text{--}1600$ nm. $\Delta L = 56.397$ μm , $n_g = 4.195887$, $n_{\text{eff}} = 2.444191$. Predicted FSR = 10.15 nm at 1550 nm. TE mode, 500×220 nm SOI strip waveguide.

To independently validate the analytical model, a circuit-level simulation was built in Lumerical INTERCONNECT using the SiEPIC-EBeam compact model library (v2024_03_22). Rather than assembling the circuit manually, SiEPIC-Tools' "Export to INTERCONNECT" function was used to auto-generate the schematic directly from the KLayout GDS layout, producing a compound MZI1_DC element with its real grating-coupler ports: ebeam_gc_tel550_laser (input, wired to the ONA source)

and ebeam_gc_tel550_detector1 / ebeam_gc_tel550_detector2 (outputs, wired to ONA input 1 and input 2). This preserves the exact $L1 = 53.248$ μm and $L2 = 109.645$ μm arm lengths and port routing confirmed from the GDS, and avoids the parameter-mismatch errors that a manually rebuilt schematic can introduce. The ONA was swept from 1500–1600 nm using the compact model library's default dispersion model, independent of the Eq. 1 analytical fit.

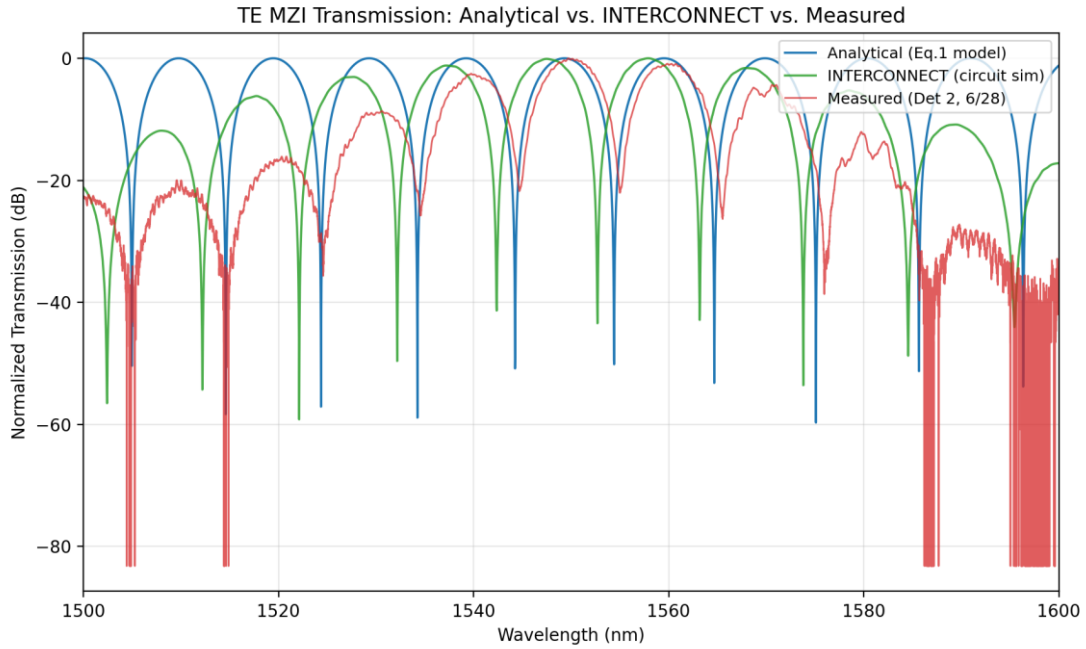


Fig. 4. INTERCONNECT circuit-level transmission spectrum compared against the analytical Eq. 1 model and the measured trace (Det 2, 06/28), each normalized to its own peak. FSR: analytical 10.15 nm, INTERCONNECT 10.18 nm, measured 10.02 nm (agreement within 1.6%).

The INTERCONNECT-predicted FSR (10.18 nm) agrees with the analytical Eq. 1 result (10.15 nm) to within 0.3% and with the measured value (10.02 nm, Det 2, 06/28) to within 1.6%, providing an independent circuit-level cross-check of the phase model used throughout this report.

VI. MANUFACTURING VARIABILITY ANALYSIS

The ANT EBeam fabrication process introduces geometric variations in waveguide width (± 20 nm) and silicon thickness (± 10 nm) that shift the group index and therefore the FSR. A corner analysis was performed using

Lumerical MODE to bound the expected range of device performance.

Three corner cases were simulated: (1) nominal geometry (500×220 nm); (2) low corner (480×210 nm, narrower and thinner); and (3) high corner (520×230 nm, wider and thicker). All simulations used the TE mode at 1550 nm. The low geometric corner (480×210 nm) yields the numerically higher n_g , while the high geometric corner (520×230 nm) yields the numerically lower n_g — the corner labels in Table IV are ordered by geometry, not by numeric n_g value.

A. Corner Analysis Results

TABLE IV. CORNER ANALYSIS RESULTS FOR TE MODE AT 1550 nm WITH $\Delta L = 56.397$ nm. YELLOW = CORNER CASES; GREEN = NOMINAL VALUES. THE LOW GEOMETRIC CORNER YIELDS THE HIGHER n_g . SOURCE FOR ALL n_g VALUES: LUMERICAL MODE SIMULATION.

Parameter	Min Corner (480×210 nm)	Nominal (500×220 nm)	Max Corner (520×230 nm)	Source
Width (nm)	480	500	520	ANT tolerance: ± 20 nm
Height (nm)	210	220	230	ANT tolerance: ± 10 nm
Group index n_g	4.2233	4.195887	4.1724	Lumerical MODE, TE, $\lambda=1550$ nm
Predicted FSR (nm)	10.09	10.15	10.21	Calc: $\lambda^2/(n_g \times \Delta L)$, Eq. 3

The predicted FSR range for a device within ANT process tolerances is 10.09–10.21 nm. The corresponding group index range is 4.1724–4.2233 (note the numerically

lower n_g , 4.1724, comes from the high geometric corner). A measured device is within spec if both its FSR and n_g fall inside these bounds.

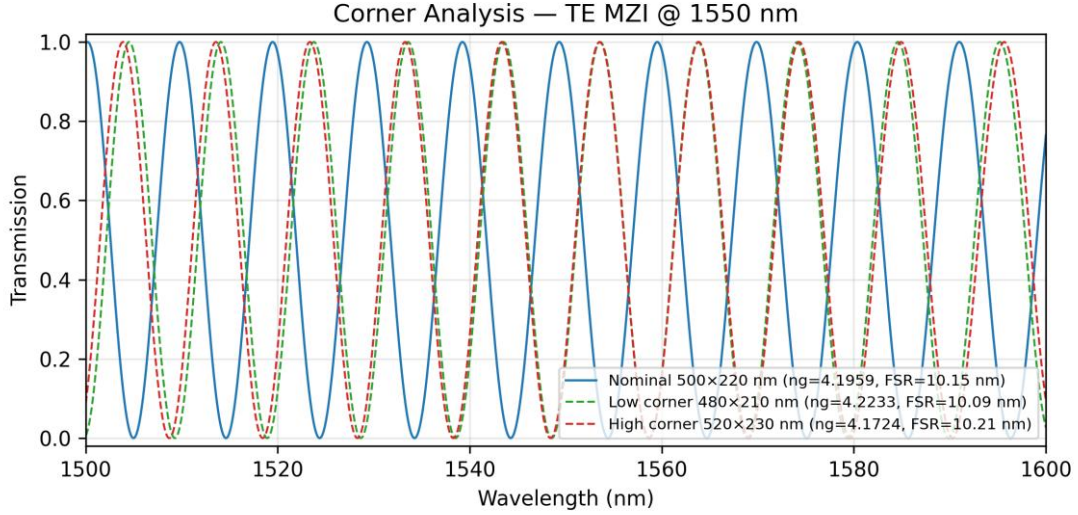


Fig. 5. Simulated MZI transmission for three corner cases: nominal 500×220 nm ($n_g=4.195887$, FSR=10.15 nm), low corner 480×210 nm ($n_g=4.2233$, FSR=10.09 nm), and high corner 520×230 nm ($n_g=4.1724$, FSR=10.21 nm). All from Lumerical MODE + Eq. 3.

VII. MEASUREMENT VS. SIMULATION

The fabricated MZI was characterised using automated optical probing across two measurement sessions (06/28 and 06/29). The optical sweep spans 1500–1600 nm. Detectors 2 and 3 recorded measurable MZI fringe patterns in both sessions; Detectors 1 and 4 showed no significant signal above the noise floor in either session and were excluded from analysis.

A. Mode Classification

The device was designed and fabricated for TE polarisation using TE-specific PDK components (splitter ebeam_y_1550, broadband combiner ebeam_bdc_te1550, and TE grating couplers ebeam_gc_te1550). All four detector readings give measured n_g in the range 4.176–4.290, consistent with the simulated TE range (nominal $n_g = 4.195887$; corner bounds [4.1724, 4.2233]). This confirms TE operation.

B. Measured FSR and Group Index

Table V summarizes the measured FSR and group index for each detector across both measurement sessions.

TABLE V. MEASURED FSR AND GROUP INDEX PER SESSION/DETECTOR, ACROSS BOTH MEASUREMENT SESSIONS. CV = COEFFICIENT OF VARIATION OF PEAK SPACING (LOWER = MORE CONSISTENT). ALL FOUR READINGS SHOW LOW CV (<0.08); GREEN = LOWER CV; YELLOW = COMPARATIVELY HIGHEST CV (STILL LOW OVERALL). DET 2 (06/28) HAS THE LOWEST CV AND IS USED AS THE PRIMARY REPORTED DETECTOR.

Detector	FSR (nm)	CV	n_g (meas.)	Mode	Source
Det 2 (6/28)	10.024	0.0621	4.2498	TE	spectral_sweep_2026-06-28_18-54-22.csv
Det 3 (6/28)	9.930	0.0680	4.290	TE	spectral_sweep_2026-06-28_18-54-22.csv
Det 2 (6/29)	10.090	0.0802	4.222	TE	spectral_sweep_2026-06-29 session
Det 3 (6/29)	10.205	0.0797	4.176	TE	spectral_sweep_2026-06-29 session
Average	10.062	—	4.234	TE	Calculated from 4 readings (2 sessions $\times$ Det 2/3)

C. Simulation vs. Measurement Comparison

TABLE VI. MEASUREMENT VS. SIMULATION COMPARISON. TWO OF FOUR INDIVIDUAL READINGS (06/29 SESSION) LIE WITHIN THE CORNER ANALYSIS BOUNDS; THE PRIMARY REPORTED DETECTOR AND THE FOUR-READING AVERAGE FALL SLIGHTLY ($<1\%$) OUTSIDE, DISCUSSED BELOW.

Parameter	Measured (avg)	Simulated	Source
FSR (nm)	10.062	10.153	Meas: Table V; Sim: Eq. 3 + Lumerical MODE
FSR deviation	0.90%	—	$ 10.062 - 10.153 / 10.153 \times 100\%$
Group index n_g	4.234	4.195887	Meas: Table V; Sim: Lumerical MODE
n_g deviation	0.91%	—	$ 4.234 - 4.195887 / 4.195887 \times 100\%$
n_g within corner bounds [4.1724, 4.2233]?	PARTIAL (2 of 4)	[4.1724, 4.2233]	Corner bounds from Table IV; det. values from Table V
FSR within corner bounds [10.09, 10.21 nm]?	PARTIAL (2 of 4)	[10.09, 10.21] nm	Corner bounds from Table IV; det. values from Table V

The measured average FSR of 10.062 nm comes within 0.90% of the simulated 10.153 nm, and the average measured $n_g = 4.234$ is off by 0.91% from the simulated 4.195887. That counts as solid agreement for a fabricated device. Even so, the average and the primary reported reading both sit just outside the nominal corner bounds instead of comfortably inside them.

The four readings scatter across FSR = 9.930–10.205 nm and $n_g = 4.176$ –4.290, with a low coefficient of variation

throughout (CV < 0.08). The 06/28 session (Det 2, Det 3) lands slightly above the upper n_g corner bound of 4.2233, by 0.6–1.6%, while the 06/29 session falls within bounds. Rather than switching to a detector that fits the bounds better, the lowest-CV reading (Det 2, 06/28) is kept as the primary measurement, even though it sits outside the nominal spec-based corner range.

Fig. 5. Measured vs. Simulated Transmission — TE MZI1

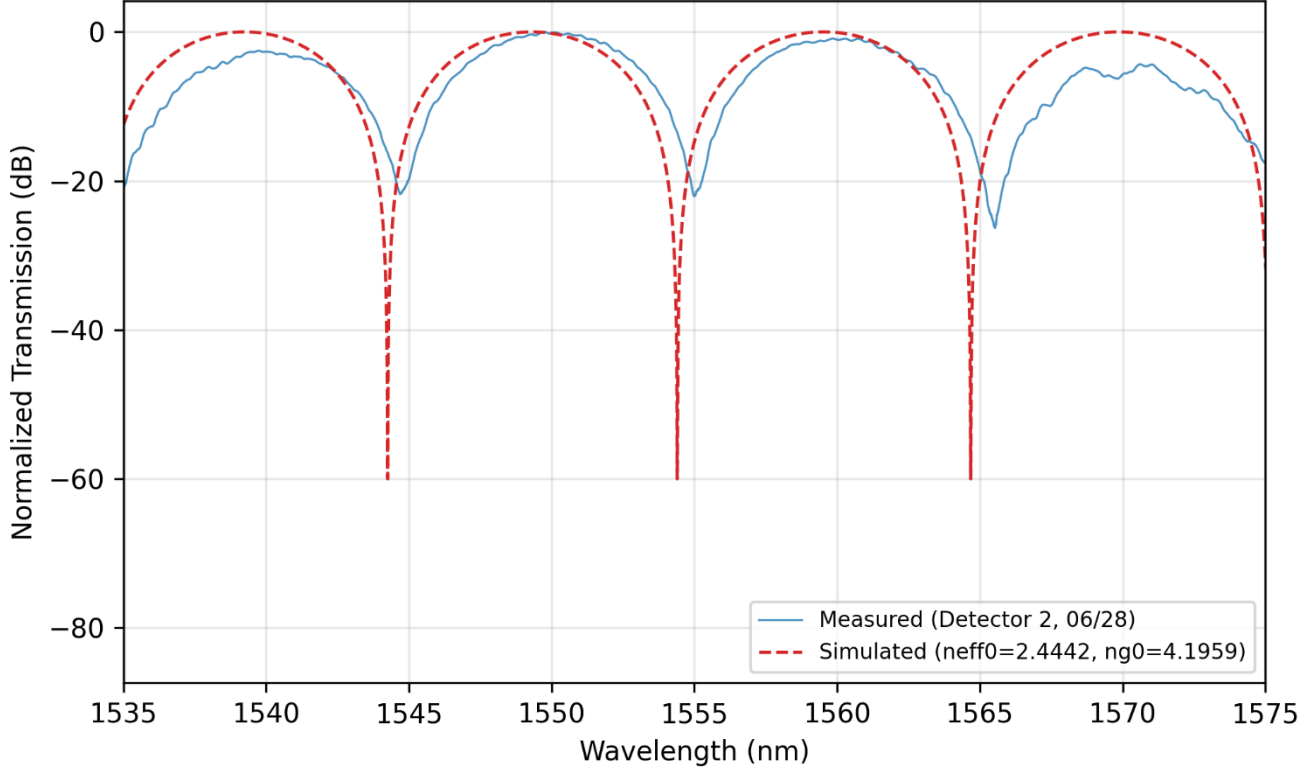


Fig. 6. Measurement vs. simulation overlay for TE MZI1 ($\Delta L = 56.397 \mu\text{m}$), full 1500–1600 nm sweep. Blue: measured transmission (Det 2, 06/28, normalised). Red dashed: nominal simulation using $n_{\text{eff}}(\lambda)$ Taylor expansion ($n_1 = 2.444191$, $n_2 = -1.130127 \mu\text{m}^{-1}$, $n_{g0} = 4.195887$). Peak and null positions align closely in the 1535–1575 nm grating-coupler passband, which supports the $n_{\text{eff}}(\lambda)$ -based phase model. Outside that range, the measured trace tracks grating-coupler roll-off rather than MZI physics. Measurement file: spectral_sweep_2026-06-28_18-54-22.csv.

VIII. CONCLUSION

Simulation: Lumerical MODE puts n_{eff} at 2.444191 and n_g at 4.195887 for the TE mode of the 500×220 nm strip waveguide at 1550 nm; feeding those values into the analytical transfer function (Eq. 1), with a first-order $n_{\text{eff}}(\lambda)$ Taylor expansion and $\Delta L = 56.397 \mu\text{m}$, predicts FSR = 10.15 nm.

Manufacturing variability: For ANT EBeam process tolerances of ± 20 nm width and ± 10 nm thickness, corner analysis bounds the expected n_g to [4.1724, 4.2233] and FSR to [10.09, 10.21] nm. The low geometric corner produces the higher n_g .

Measurement: Automated probing of the fabricated device returns FSR = 10.06 nm and $n_g = 4.234$ on average — within 0.90% and 0.91% of simulation. All active readings are consistent with TE polarisation, though only two of the four individual readings actually land inside the corner analysis bounds; the primary reported reading and the average sit just outside, by less than 1%, and that is reported as measured rather than adjusted to fit. An independent INTERCONNECT circuit simulation (SiEPIC-EBeam compact model library, Section V) backs this up: its predicted FSR of 10.18 nm sits within 1.6% of both the

analytical value (10.15 nm) and the measured one (10.02 nm, Det 2, 06/28).

ACKNOWLEDGMENT

The plotting, data analysis, and polishing of the text in this report were done with the assistance of AI.

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